Reflective Report: Ethical, Social and Professional Impacts of Cloud Computing at Amani Bank Ltd

How cloud technologies shape decision-making in a small-bank scenario

Setting the scene

When Amani Bank Ltd, a small Kenyan bank, asked me to help move its mobile banking, disaster recovery and analytics to a hybrid cloud, I assumed the hard part would be technical. I started by separating the service models, infrastructure, platform and software as a service, and the deployment options of public, private and hybrid cloud (Mell and Grance, 2011). Choosing hybrid, with the core ledger kept on premises, felt sensible. I quickly learned that the decisions that mattered most were not technical.

Ethical impacts

The first discomfort was data. Customer records are not just assets; they belong to people who trust the bank. Once data sits with a provider, questions of control, jurisdiction and secondary use become real (Hon, Millard and Walden, 2011), and Zuboff (2019) reminds us how easily data is turned into commercial advantage. Kenya's Data Protection Act requires safeguards for personal data and for cross-border transfers (Republic of Kenya, 2019), so I chose in-country storage for regulated data and customer-managed encryption keys. There is also an environmental dimension: data centres consume

substantial energy (Masanet et al., 2020), so I now treat efficient sizing and region choice as ethical decisions, not just cost ones.

Social impacts

Cloud lowers entry costs, which helps smaller institutions serve customers previously excluded from banking (Armbrust et al., 2010). Yet it also changes jobs. Colleagues who managed servers feared redundancy, and I realised that reskilling in cloud operations and security is part of a responsible migration. Dependence matters too: if a provider fails, branches and mobile customers in several countries are affected together. Fairness is also a concern, because customers with poor connectivity may be the first to suffer when digital channels degrade, so resilience is partly a question of inclusion.

Professional impacts

Professionally, outsourcing a service does not outsource accountability. CBK outsourcing and cybersecurity guidance expects board oversight, audit rights and exit plans (Central Bank of Kenya, 2013; 2017). I relied on ISO/IEC 27017 and 27018 for cloud controls and the protection of personal information (ISO/IEC, 2015; 2019), and on the ACM Code of Ethics (ACM, 2018) as a reminder to put public interest ahead of convenience. Configuring identity, logging and backup myself showed me how a single misconfiguration can undermine a good design, which is why I favour automated guardrails and tested recovery over heavy duplication.

Influence on decisions

Together, these impacts changed what we decided. We kept regulated data local, put exit clauses into contracts, funded training before go-live, and phased the migration so critical services had tested failover. Cheapest and fastest were no longer the deciding criteria; defensible and trustworthy became the test we applied to every option.

Lessons learnt

Using Gibbs' (1988) cycle, I felt anxious at first, then realised I had under-weighted people and regulation. I learned that governance is a design input, that trust is the real currency of a bank, and that resilience should be proportionate. Next time I would involve compliance, legal and staff representatives earlier and rehearse a provider-failure scenario before committing. Ultimately, cloud is a tool whose value depends on the judgement of the people who govern it.

Learning outcomes addressed

Learning outcome	Where it is addressed
1. Fundamental concepts of cloud computing	Setting the scene: service and deployment models (Mell and Grance, 2011)
2. Different cloud solutions	Hybrid choice with the core ledger on premises; public cloud for channels, recovery and analytics

Learning outcome	Where it is addressed
3. Configure and implement cloud services and tools	Professional impacts: identity, logging, backup and automated guardrails
4. Methodologies for resilient solutions	Influence on decisions: phased migration, tested failover, proportionate resilience
5. Standards and procedures	ISO/IEC 27017 and 27018, CBK guidance, Data Protection Act 2019, ACM Code of Ethics

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